THE DERIVED PULLBACK PRESERVES THE CONSTRUCTIBLE FINITE TYPE DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction through the characterisation criterion and the two inputs (exactness of the pullback and constructibility of pullbacks) that give the conclusion.

Editorial note. The label lemma-pullback-ctf names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The criterion describing the constructible finite type category. Same chapter, the lemma characterising membership in the constructible finite type derived category through finite complexes of flat constructible sheaves of Lambda-modules. Supplies the reduction of the statement to finite complexes of flat constructible sheaves.
- Pullback of constructible sheaves. Same chapter, the lemma stating that the pullback of a constructible sheaf of Lambda-modules is constructible. Supplies constructibility of the pullback of each term.

Source reference. Original reference 095G: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11808–11808. Prerequisite pointer invoked at proof line(s) 13846; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03TT: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 13744–13744. Prerequisite pointer invoked at proof line(s) 13843; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let Λ be a Noetherian ring. Let $f : X \to Y$ be a morphism of schemes. If $K \in D_{ctf}(Y_{étale}, \Lambda)$ then $Lf^*K \in D_{ctf}(X_{étale}, \Lambda)$.*

Proof. Apply Lemma 03TT to reduce this to a question about finite complexes of flat constructible sheaves of Λ-modules. Then the statement follows as $f^{-1} = f^*$ is exact and Lemma 095G. □

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